An NLP-Driven System for Automated Study Material Generation and Continuous Assessment Guidance

Project ID: 25-26J-089

Project Proposal Report

Janushiya R – IT22171306

B.Sc. (Hons) Degree in Information Technology Specializing in Data Science

Department of Information Technology

Sri Lanka Institute of Information Technology

Sri Lanka

August-2025

An NLP-Driven System for Automated Study Material Generation and Continuous Assessment Guidance

Project ID: 25-26J-089

Project Proposal Report

Janushiya R – IT22171306

B.Sc. (Hons) Degree in Information Technology Specializing in Data Science

Department of Information Technology

Sri Lanka Institute of Information Technology

Sri Lanka

August-2025

Declaration

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

Name	Student ID	Signature	Date
Janushiya R	IT22171306		30/08/2025

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

Signature of the supervisor:

(Dr. Sanvitha Kasthuriarachchii) Date----------

Signature of the Co-supervisor:

(Ms. Karthiga Rajendran) Date----------

Abstract

This research proposes the development of a domain-specific AI-powered academic support system aimed at improving the self-learning process of university students by converting unstructured educational resources such as lecture slides and continuous assessment (CA) questions from the Database Management Systems (DMS) module into structured, syllabus-aligned, and personalized study aids. The solution is designed around a Retrieval-Augmented Fine-Tuning (RAFT) framework that combines Retrieval-Augmented Generation (RAG) with Reinforcement Learning from Human Feedback (RLHF), enabling the system to continuously refine its outputs based on student interactions and feedback. Lecture slides are summarized using advanced Large Language Models (LLMs) enhanced with RAFT, producing concise and conceptually accurate notes which are further transformed into multimodal learning formats, including revision-mode and concept-mode voice notes as well as Bloom's Taxonomy–based adaptive flashcards that balance easy, medium, and hard questions. For continuous assessment support, an Agentic RAG pipeline integrated with syllabus-specific datasets and a knowledge graph provides step-by-step guidance, verified solutions, and curated academic references, ensuring accuracy, explainability, and relevance to institutional standards. By combining LLMs, RAG, RAFT, RLHF, Knowledge Graphs, and Multimodal AI, the proposed system delivers a scalable, syllabus-aware, and adaptive academic support platform that enhances student engagement, conceptual clarity, exam readiness, and overall learning efficiency.

Keywords - Retrieval-Augmented Generation (RAG), Retrieval-Augmented Fine-Tuning (RAFT), Reinforcement Learning from Human Feedback (RLHF), Large Language Models (LLMs), Knowledge Graph, Bloom's Taxonomy, multimodal learning, lecture summarization, adaptive flashcards, text-to-speech (TTS), continuous assessment (CA) guidance, syllabus alignment, Database Management Systems (DMS).

Contents

1. Introduction 1

1.1 Background & Literature Survey 2

1.2 Research Gap 4

1.3 Research Problem 5

2. Objectives 7

2.1 Main Objectives 7

2.2 Specific Objectives 7

3. Methodology 9

3.1 Requirement Gathering 9

3.2 Past Research Analysis 9

3.3 Identifying Existing System 11

3.4 Feasibility Study 12

3.4.1 Technical Feasibility Study 12

3.4.2 Schedule Feasibility 12

3.4.3 Economic Feasibility 13

3.5 System Analysis 14

3.5.1 System Solution Approach 14

3.5.2 Tools and Technology 15

3.6 Overall System Diagram 17

3.7 Project Requirements 20

3.7.1 Functional Requirements 20

3.7.2 Non-Functional Requirements 21

3.8 Testing 23

3.9 Timeline 25

3.10 Risk Management Plan 27

3.11 Communication Management Plan 29

4. Commercialization 31

5. Budget 32

6. Conclusion 33

7. References 34

List of Figures

Figure 1- Jira dashboard showing project tasks and progress tracking..... 13

Figure 2- Component Flow Diagram..... 17

Figure 3- Component Diagram 18

Figure 4- System Diagram 19

Figure 5- Project Grantt Chart..... 26

List of Tables

Table 1-Comparative analysis of existing research 5

Table 2 - Tools and Technologies 16

Table 3- Project Timeline 26

Table 4- Risk Identification 28

Table 5- Budget for Component 32

List of Abbreviations

AI	Artificial Intelligence
NLP	Natural Language Processing
LLM	Large Language Model
TTS	Text-to-Speech
RAG	Retrieval-Augmented Generation
RAFT	Retrieval-Augmented Fine-Tuning
RLHF	Reinforcement Learning from Human Feedback
KG	Knowledge Graph
DMS	Database Management Systems
CA	Continuous Assessment
TTS	Text-to-Speech
IDE	Integrated Development Environment
VCS	Version Control System
UAT	User Acceptance Testing

UI	User Interface
API	Application Programming Interface

1. Introduction

Assessment is one of the most influential drivers of student learning in higher education, yet many universities struggle to maintain consistent alignment between curriculum delivery, lecture content, and examination practices. This misalignment often results in students devoting significant time to topics that appear only rarely in exams, while overlooking those that are frequently tested and more critical for success. As a consequence, students' preparation becomes inefficient and learning outcomes are not fully achieved. Traditional approaches, such as manually analyzing past exam papers and lecture materials, are often subjective, inconsistent, and time-consuming, making them inadequate for addressing this issue on a larger scale.

The emergence of Artificial Intelligence (AI), Natural Language Processing (NLP), and educational analytics presents a timely opportunity to transform this process. Unlike conventional learning management systems, which largely focus on content delivery and administrative functions, AI-powered approaches can uncover structural and topical trends across years of academic assessments, identify gaps between taught material and examined content, and provide actionable insights for both learners and educators. These capabilities not only optimize revision strategies for students but also guide institutions in improving curriculum design and instructional effectiveness.

Our research targets the development of an intelligent, AI-powered academic support framework that systematically analyzes historical exam papers, lecture content, and continuous assessments to deliver data-driven insights and personalized learning outputs. Through advanced NLP techniques, the system identifies high-frequency topics, predicts question difficulty, and classifies content according to syllabus requirements, while also highlighting redundancies and underrepresented areas. Based on these analyses, the framework generates practical learning aids, including model exam papers that reflect historical patterns, concise and curriculum-aligned short notes, adaptive essay evaluation with feedback, interactive flashcards, voice notes, and personalized time-aware study schedules.

By integrating these elements into a unified platform, the research provides a holistic solution that enhances student self-learning, improves exam preparation efficiency, and strengthens curriculum

alignment with assessment practices. Ultimately, the project seeks to bridge the gap between academic delivery and evaluation through a data-driven, scalable, and personalized educational support system, thereby contributing to improved student achievement and institutional teaching quality.

1.1 Background & Literature Survey

The rapid integration of Artificial Intelligence (AI) into the education sector has revolutionized how students interact with academic materials, shifting from purely instructor-led learning to more interactive and technology-enhanced self-learning experiences. In higher education, this shift is particularly relevant to technical modules such as Database Management Systems (DMS), where students must acquire both theoretical knowledge and hands-on skills through lectures, tutorials, laboratory sessions, and continuous assessments (CA). Traditionally, revision for such courses relies on manually going through lecture slides, creating personal notes, and searching for additional reference materials from textbooks or online sources. This process is often inefficient and inconsistent, especially when students have limited time to prepare for CA tasks that require practical problem-solving skills. Without targeted learning aids, students may focus on less relevant content, overlook key concepts, or fail to connect theoretical knowledge with assessment requirements. These challenges highlight the growing demand for academic support tools that can automatically process lecture materials, structure them into concise and easily digestible formats, and provide guidance that aligns directly with syllabus objectives.

Recent advances in Large Language Models (LLMs), also referred to as foundation models, have transformed educational AI by moving beyond task-specific NLP toward flexible systems capable of summarization, explanation, and problem-solving with minimal supervision [1]. However, unconstrained generation risks factual inaccuracies and misalignment with course objectives. To address this, Retrieval-Augmented Generation (RAG) has been introduced, conditioning outputs on relevant retrieved sources to improve accuracy and reliability in knowledge-intensive tasks [2]. This approach has been further extended into Retrieval-Augmented Fine-Tuning (RAFT), which integrates

retrieval with fine-tuning strategies guided by Reinforcement Learning from Human Feedback (RLHF). RLHF has been shown to align models with human preferences by combining reinforcement learning with human-evaluated outputs, both in large-scale instruction tuning [3] and in foundational preference-based reinforcement learning studies [4]. Furthermore, retrieval-aware fine-tuning enhances few-shot generalization by teaching models to effectively incorporate retrieved evidence [5].

In parallel, the use of Knowledge Graphs (KGs) has proven valuable in educational contexts, enabling structured representation of syllabus concepts and prerequisite relationships. By integrating KGs with RAG, educational tutoring systems achieve improved retrieval precision, conceptual coverage, and factual verification [6]. Alongside knowledge-grounded methods, multimodal learning has been widely recognized as an effective way to enhance comprehension and retention. Mayer's principles of multimedia learning highlight that presenting information across multiple modalities, such as text combined with audio narration, improves accessibility and supports diverse learner needs [7].

Another important area of educational research is active recall, where flashcards remain a proven technique for long-term retention. When structured around Bloom's Taxonomy, flashcards can support progressive cognitive development: *remembering* factual definitions, *understanding* conceptual differences, *applying* knowledge to problem-solving scenarios, *analyzing* relationships, *evaluating* correctness, and eventually *creating* new solutions [8]. Recent AI-enabled assistants in higher education have operationalized this by generating quizzes and flashcards automatically, with adaptivity driven by learner performance and feedback [9]. When coupled with RLHF, this adaptivity allows flashcard generation to adjust difficulty and style dynamically, aligning with student needs and syllabus objectives.

Although Intelligent Tutoring Systems (ITS) have long demonstrated the benefits of guided learning and adaptivity, many remain generic, open-domain, or static, lacking integration of retrieval grounding, reinforcement-based alignment, knowledge graph verification, multimodal delivery, and Bloom's-aligned adaptive learning [10]. This gap highlights the research opportunity to develop a domain-specific academic support platform that leverages LLMs, RAG/RAFT, RLHF, KGs, multimodal

delivery, and Bloom's Taxonomy to transform unstructured Database Management Systems materials into structured, syllabus-aware, and continuously improving learning aids.

1.2 Research Gap

Despite notable progress in educational AI, existing systems remain fragmented and limited in scope. Most current tools emphasize either text summarization or generic question-answering, but fail to provide an integrated framework that combines syllabus-aware summarization, multimodal delivery, adaptive flashcards, and continuous assessment guidance within a single platform. While Retrieval-Augmented Generation (RAG) has improved factual grounding, few systems extend this into Retrieval-Augmented Fine-Tuning (RAFT) with Reinforcement Learning from Human Feedback (RLHF), which is essential for creating adaptive, feedback-driven learning resources. Similarly, although Knowledge Graphs (KGs) have shown promise in representing conceptual hierarchies, they are rarely applied to enforce syllabus coverage and correctness in higher education modules such as Database Management Systems (DMS). Furthermore, existing intelligent tutoring systems are often generic, open-domain, and static, lacking personalization and Bloom's Taxonomy–aligned progression from remembering to applying and analyzing. These gaps highlight the need for a domain-specific, syllabus-aligned, and continuously improving academic support platform that integrates LLMs, RAFT/RLHF, KGs, multimodal delivery, and Bloom's-based adaptive flashcards to effectively bridge lecture-based learning with continuous assessment preparation.

Feature	Research 01 [7]	Research 02 [8]	Research 03 [9]	Proposed System
Lecture summarization	✗	✗	✓	✓

Generate flashcards	✗	✓	✓	✓
Multimodal delivery (voice notes, flashcards)	✓	✗	✗	✓
Syllabus-aware CA guidance	✗	✗	✗	✓
Integration of RAFT + RLHF	✗	✗	✗	✓
Knowledge Graph support	✗	✗	✗	✓
Bloom's Taxonomy–based adaptivity	✗	✓	✗	✓

Table 1-Comparative analysis of existing research

1.3 Research Problem

Students in the Database Management Systems (DMS) module face challenges in efficiently revising lecture material, reinforcing key concepts, and preparing for continuous assessments (CA) in a structured and syllabus-focused manner. The absence of an integrated platform that can generate concise summaries, provide multimodal study aids, create adaptive flashcards, and deliver syllabus-aligned CA guidance results in fragmented and inefficient learning. Existing tools either

focus narrowly on text summarization or offer generic question-answering, but they lack syllabus awareness, adaptivity, and integration across different learning formats.

This research seeks to address these limitations by developing a domain-specific, multimodal AI-powered academic support system that combines Large Language Models (LLMs), Retrieval-Augmented Fine-Tuning (RAFT), Reinforcement Learning from Human Feedback (RLHF), Knowledge Graphs (KGs), and Bloom's Taxonomy–based adaptivity. Lecture slides will be transformed into concise, syllabus-aligned summaries, which are then delivered in both text and audio formats. These summaries will also serve as the basis for generating Bloom's-aligned flashcards that adapt to student performance and feedback. For assessment preparation, the system employs an Agentic RAG pipeline that integrates syllabus-specific datasets and knowledge graphs to provide step-by-step CA guidance and verified solutions. By aligning outputs directly with the syllabus and incorporating continuous feedback, the system aims to enhance student engagement, improve conceptual clarity, and provide adaptive, syllabus-aware learning support for academic success.

2. Objectives

2.1 Main Objectives

To design and implement a domain-specific, multimodal AI-powered academic support system for the Database Management Systems (DMS) module that leverages Large Language Models (LLMs), Retrieval-Augmented Fine-Tuning (RAFT), Reinforcement Learning from Human Feedback (RLHF), and Knowledge Graphs (KGs) to transform lecture slides into concise, syllabus-aligned summaries, generate Bloom's Taxonomy–based adaptive flashcards, deliver multimodal study aids such as voice notes, and provide step-by-step, syllabus-aware continuous assessment (CA) guidance.

2.2 Specific Objectives

1. **Data Acquisition and Preprocessing**: Collect, clean, and preprocess Database Management Systems (DMS) lecture slides and continuous assessment (CA) datasets in order to prepare them for embedding, retrieval, and model training.

2. **Lecture Summarization**: Implement a summarization pipeline using Large Language Models (LLMs) enhanced with Retrieval-Augmented Fine-Tuning (RAFT) to generate concise, syllabus-aligned summaries from lecture content.

3. **Flashcard Generation**: Develop an intelligent flashcard generation module capable of producing flashcards structured according to Bloom's Taxonomy levels of remembering, understanding, applying, analyzing, evaluating, and creating. Reinforcement Learning from Human Feedback (RLHF) will be incorporated to adapt difficulty and style based on student performance and feedback.

4. **Voice Note Creation**: Integrate a Text-to-Speech (TTS) system to convert lecture summaries into audio study notes. Two delivery formats will be supported: revision mode, which provides concise bullet-style narration, and concept mode, which delivers detailed explanations with examples.

5. **Continuous Assessment (CA) Problem-Solving Guidance**: Design and implement an Agentic Retrieval-Augmented Generation (RAG) pipeline supported by Knowledge Graphs to retrieve relevant syllabus content and provide step-by-step, syllabus-aware CA solutions, including verified explanations and curated references.

6. **Syllabus Mapping and Knowledge Graph Integration**: Build and integrate a Knowledge Graph (KG) to ensure that all outputs, including summaries, flashcards, voice notes, and CA guidance, are consistently aligned with the official DMS syllabus and structured concept dependencies.

3. Methodology

3.1 Requirement Gathering

The requirement gathering stage plays a vital role in shaping this component. It begins with a thorough review of recent studies to examine existing approaches to lecture summarization, flashcard generation, and CA support. To maintain reliability and syllabus relevance, lecture slides and CA-related datasets will be sourced from officially approved academic materials. These resources will then be studied to identify the main difficulties students encounter in grasping and retaining DMS content. Guidance from subject lecturers will also be incorporated to ensure that the outputs produced summaries, flashcards, voice notes, and CA guidance are consistent with the official syllabus and uphold academic standards. Through this process, the component's requirements are clearly defined, balancing both educational needs and technical feasibility to produce an effective academic support tool.

3.2 Past Research Analysis

Research on educational AI has evolved significantly with the rise of Large Language Models (LLMs), which extend beyond task-specific NLP architectures to provide flexible, general-purpose summarization, question answering, and reasoning capabilities [1]. While these models demonstrate impressive performance, unconstrained generation can lead to factual drift and lack of syllabus alignment. To address these limitations, Retrieval-Augmented Generation (RAG) was introduced, grounding outputs in retrieved evidence to enhance accuracy and contextual reliability in knowledge-intensive tasks [2]. Building on this foundation, recent advancements in Retrieval-Augmented Fine-Tuning (RAFT) integrate retrieval mechanisms with Reinforcement Learning from Human Feedback (RLHF), enabling systems to adapt continuously through user interactions and feedback [3][4][5]. These developments provide the methodological basis for adaptive summarization and assessment support that improves over time in response to student feedback.

Another research direction has emphasized the use of Knowledge Graphs (KGs) to represent structured academic content. By encoding conceptual hierarchies and prerequisite relationships, KGs have been shown to improve retrieval precision, syllabus coverage, and factual verification when combined with RAG-based tutoring frameworks [6]. This supports the integration of KGs into academic support systems for modules such as DMS, where dependencies between topics (e.g., normalization, relational algebra, SQL joins) are crucial for accurate guidance.

The role of multimodal learning has also been highlighted in prior studies. Mayer's work on multimedia learning shows that students retain information more effectively when text is paired with narration or other modalities [7]. This aligns with the integration of Text-to-Speech (TTS) voice notes alongside summaries and flashcards, ensuring accessibility for auditory learners and flexibility for on-the-go revision.

In addition, Bloom's Taxonomy provides an established pedagogical framework for structuring learning progression across cognitive levels, from remembering and understanding to applying, analyzing, evaluating, and creating [8]. This taxonomy has informed recent AI-enabled assistants that generate quizzes and flashcards for higher education, with systems such as those described by Banerjee et al. [9] demonstrating how adaptive learning aids can enhance retention and engagement. When coupled with RLHF, Bloom's-aligned flashcards become more dynamic, adjusting difficulty and depth according to student performance.

Finally, the broader field of Intelligent Tutoring Systems (ITS) has long demonstrated the value of guided, adaptive learning environments. However, as Nkambou et al. [10] observe, many existing ITS solutions remain generic or static, lacking tight integration with retrieval-augmented models, reinforcement-driven alignment, multimodal delivery, and syllabus-specific knowledge structures.

Taken together, these studies establish the foundation for the proposed system. Prior research validates the use of LLMs, RAG/RAFT, RLHF, Knowledge Graphs, multimodal narration, and Bloom's Taxonomy in education, but no existing work has combined these approaches into a single, domain-specific academic support platform. This underscores the novelty of developing a syllabus-aligned, multimodal, and feedback-driven system tailored to the DMS module that delivers concise

summaries, adaptive flashcards, accessible voice notes, and step-by-step continuous assessment guidance.

3.3 Identifying Existing System

In reviewing existing educational support platforms, several limitations were identified in relation to the objectives of this study. Popular tools such as Quizlet provide flashcard-based learning but rely on manual entry of content, lacking automation, syllabus awareness, and adaptivity. Online summarization platforms can generate shortened text but are not optimized for lecture slides, do not preserve topic flow, and lack mechanisms to ensure syllabus coverage. Similarly, voice note applications exist as standalone tools, but they are not integrated with summarization or flashcard features, which limits their effectiveness as comprehensive learning aids.

With respect to Continuous Assessment (CA) support, most current systems provide only generic explanations or solutions that are unrelated to institutional datasets. They do not incorporate retrieval-augmented methods or knowledge graph verification to ensure context-aware, syllabus-specific, and factually aligned guidance. Furthermore, existing e-learning systems generally operate in isolation, offering single features such as quizzes, flashcards, or TTS, but without reinforcement-driven personalization or Bloom's Taxonomy–aligned adaptivity.

This review of existing systems underscores the need for a unified, domain-specific AI-powered solution that combines LLM-based summarization, multimodal study aids, Bloom's-aligned adaptive flashcards, and Agentic RAG–driven CA guidance into a single platform. By integrating RAG/RAFT, RLHF, and Knowledge Graphs, the proposed system addresses gaps in automation, personalization, syllabus alignment, and continuous improvement that are not currently met by existing solutions.

3.4 Feasibility Study

3.4.1 Technical Feasibility Study

3.4.1.1 Knowledge on Technologies

The solution applies Large Language Models (LLMs) enhanced with Retrieval-Augmented Fine-Tuning (RAFT) for lecture summarization, while Reinforcement Learning from Human Feedback (RLHF) enables adaptive improvement from student feedback. An Agentic RAG pipeline combined with Knowledge Graphs (KGs) supports accurate CA guidance. For multimodal delivery, Text-to-Speech (TTS) engines generate voice notes, ensuring accessibility.

3.4.1.2 Knowledge on Tools

Development will use open-source tools including PyMuPDF and python-docx for text extraction, SpaCy/NLTK for preprocessing, FAISS/Milvus for retrieval, and Neo4j for Knowledge Graphs. Hugging Face TRL supports RLHF, while React will be used for the user interface. These tools are well-documented, enabling efficient development and integration.

3.4.1.3 Data Collection Knowledge

Academic content including lecture slides and CA datasets will be obtained from institution-approved educational resources to ensure syllabus accuracy. Preprocessing methods such as text cleaning, tokenization, and chunking will prepare data for syllabus alignment, Bloom's-based flashcards, and adaptive refinement.

3.4.2 Schedule Feasibility

An essential factor in this project is ensuring that the schedule is both practical and achievable within the given academic timeframe. The development process follows an Agile methodology, where the work is divided into iterative sprints, each addressing a specific module such as summarization, flashcard generation, voice note creation, and CA guidance. To support efficient project management, tasks are carefully planned, assigned, and tracked using Jira, allowing for detailed

scheduling and continuous monitoring of progress. This approach not only facilitates early detection of potential delays but also enables timely adjustments and reallocation of resources when necessary. By adopting these practices, the project maintains a realistic timeline and ensures that deliverables are produced on schedule, contributing to the overall feasibility and success of the system.

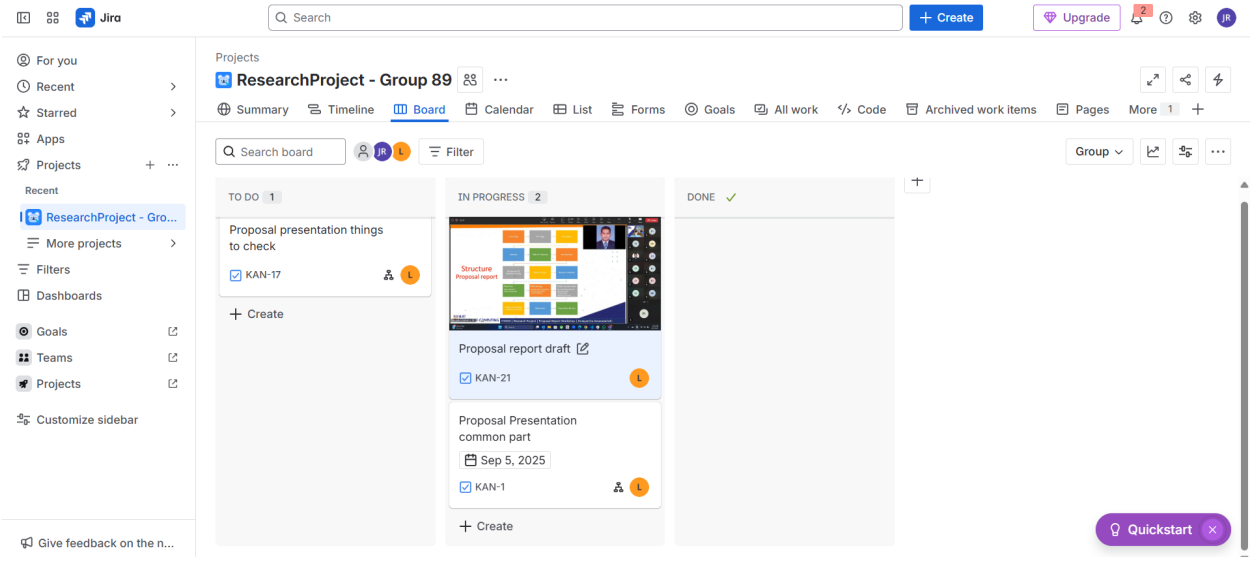

Figure 1- Jira dashboard showing project tasks and progress tracking

3.4.3 Economic Feasibility

Economic feasibility involves evaluating the potential costs associated with development, tools, and infrastructure for this component. The system primarily relies on open-source frameworks and pre-trained models, including LLMs available through Hugging Face, LangChain or LlamaIndex for retrieval pipelines, and Hugging Face TRL for reinforcement learning with human feedback. Vector databases such as FAISS or Milvus and Neo4j Community Edition for Knowledge Graphs are freely available, minimizing licensing costs. For multimodal support, Text-to-Speech engines such as gTTS and Amazon Polly (with free-tier usage) provide cost-effective options. Development platforms such

as Streamlit and React are also open-source, further reducing expenses. Infrastructure requirements are modest since the system can be developed and tested on standard computing resources, with smaller open-source LLMs ensuring efficiency. By leveraging open-source technologies and lightweight deployment strategies, the project remains cost-effective and sustainable within academic resource constraints.

3.5 System Analysis

3.5.1 System Solution Approach

The development of the proposed academic support system involves the following steps:

1. **Data Collection:** Lecture slides and continuous assessment (CA) datasets are collected from institution-approved academic sources to ensure reliability and alignment with the official syllabus.
2. **Preprocessing:** Raw text is extracted and cleaned using parsing libraries such as PyMuPDF and python-docx, with preprocessing tasks including tokenization, normalization, and formatting to prepare data for embedding and retrieval.
3. **Summarization:** Concise, syllabus-aligned summaries are generated using Large Language Models (LLMs) enhanced with Retrieval-Augmented Fine-Tuning (RAFT). This ensures summaries are both contextually accurate and adaptive.
4. **Flashcard Generation:** Summaries are transformed into structured flashcards based on Bloom's Taxonomy levels (remembering, understanding, applying, analyzing, evaluating, creating). Reinforcement Learning from Human Feedback (RLHF) is incorporated to adapt difficulty and style according to student interactions.
5. **Voice Note Creation:** Summaries are converted into audio files using Text-to-Speech (TTS) engines, providing learners with multimodal support through both revision mode (concise points) and concept mode (detailed explanations).

6. **CA Guidance:** An Agentic Retrieval-Augmented Generation (RAG) pipeline, supported by a Knowledge Graph (KG), retrieves relevant syllabus content and past CA materials to provide step-by-step, syllabus-aware solutions with verified explanations.
7. **Output Delivery:** Learners access summaries, adaptive flashcards, audio notes, and CA guidance through an interactive and user-friendly web interface developed with Streamlit or React.

3.5.2 Tools and Technology

The following tools and technologies will be used for developing the component:

Category	Tools / Technologies
Programming Languages	Python (backend, NLP, and ML pipelines), JavaScript (frontend)
Machine Learning / NLP Frameworks	Hugging Face Transformers, PyTorch, TensorFlow, LangChain, LlamaIndex (for RAFT and Agentic RAG)
Data Handling Tools	PyMuPDF, python-docx (text extraction), SpaCy, NLTK (preprocessing)
Vector Databases	FAISS, Milvus (for similarity search and retrieval)
Knowledge Graph Tools	Neo4j (concept mapping and syllabus alignment)
Data Analysis Tools	Pandas, NumPy
Web Development Frameworks	React JS, Streamlit
UI Design Implementation	Figma
Diagramming Tools	Draw.io
Integrated Development Environment (IDE)	Visual Studio Code
Version Control System (VCS)	Git (GitHub / GitLab)
Collaboration Tools	Microsoft Teams, WhatsApp

Testing Tools	UnitTest, Postman
Deployment Tools	Docker (containerization), Kubernetes (orchestration for scaling)
Project Management	Jira (sprint planning, task tracking, progress monitoring)

Table 2 - Tools and Technologies

3.6 Overall System Diagram

3.6.1 Component Flow Diagram

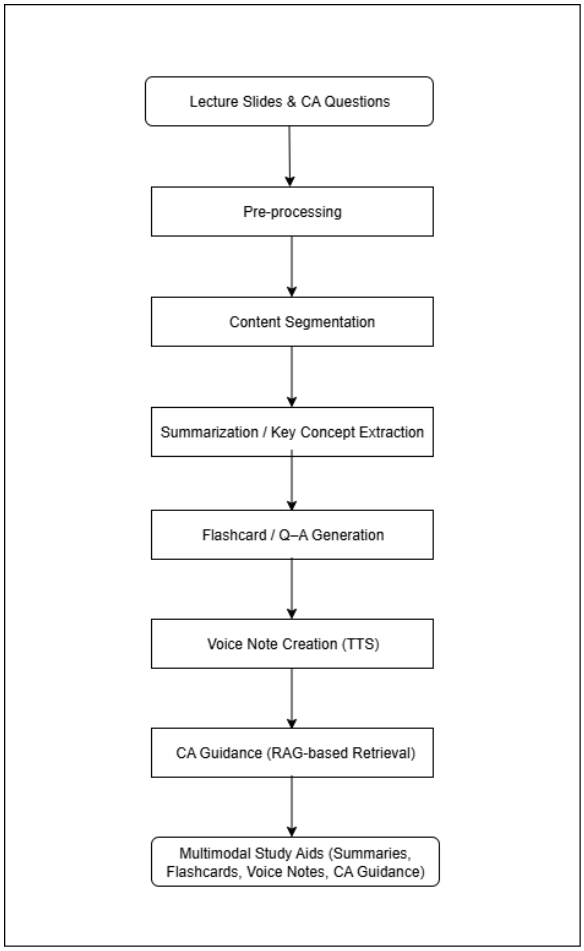

Figure 2- Component Flow Diagram

3.6.2 Component Diagram

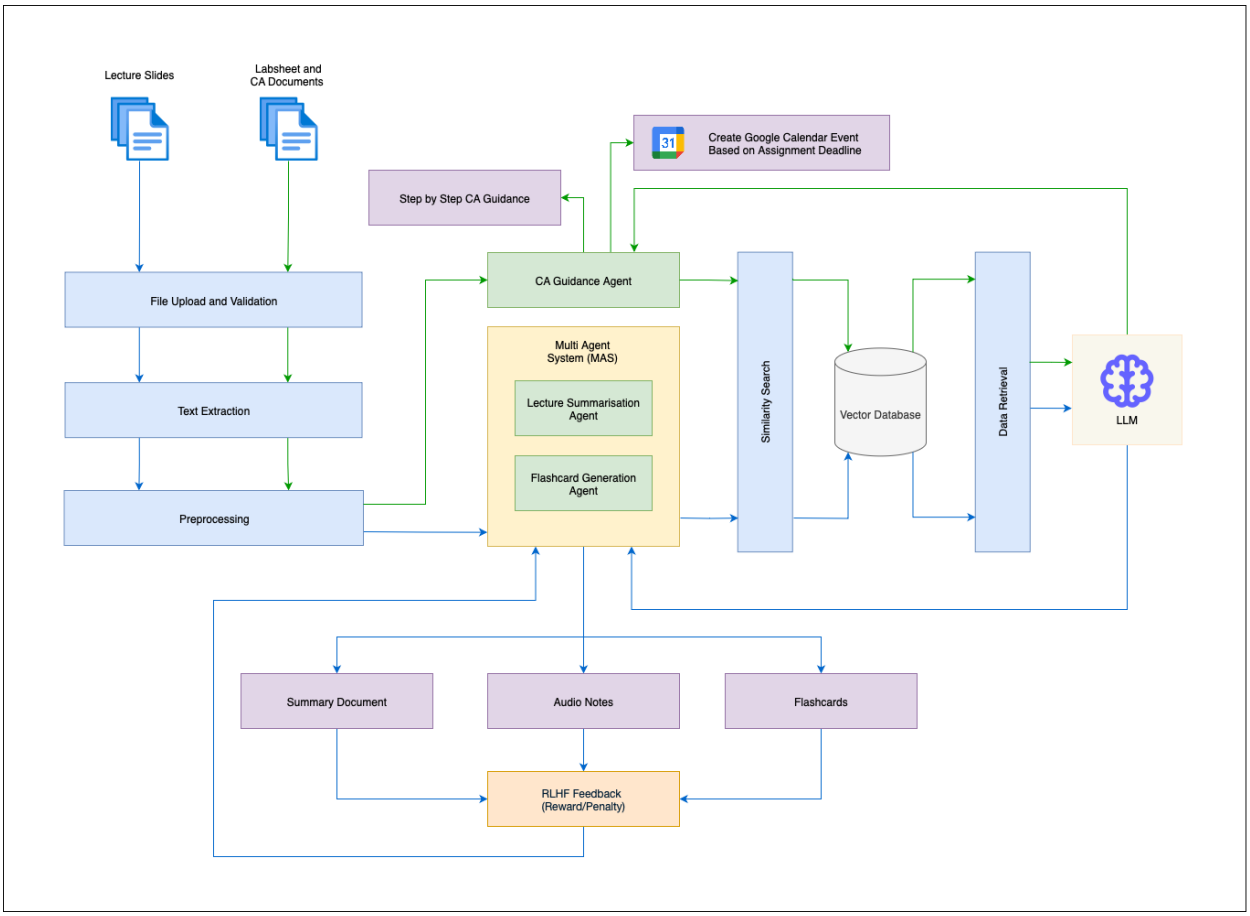

Figure 3- Component Diagram

3.6.3 System Diagram

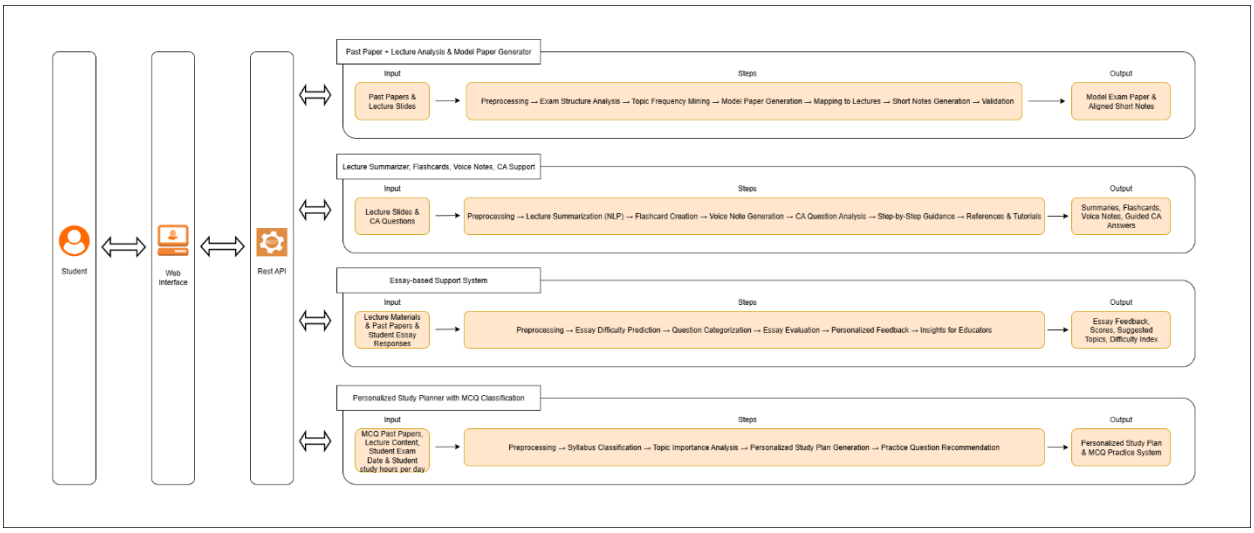

Figure 4- System Diagram

3.7 Project Requirements

3.7.1 Functional Requirements

1. Data Input and Preprocessing

- The system must accept DMS lecture slides in formats such as PDF, PPT, and DOCX.
- The system must process continuous assessment (CA) datasets from institution-specific sources.
- Perform text extraction, cleaning, tokenization, and syllabus mapping to prepare content for retrieval and model training.

2. Lecture Summarization

- Implement Large Language Models (LLMs) enhanced with Retrieval-Augmented Fine-Tuning (RAFT) to generate concise and syllabus-aligned summaries.
- Summaries must preserve the conceptual flow of lectures and emphasize high-priority syllabus topics.

3. Flashcard Generation

- Generate adaptive flashcards from summarized content based on Bloom's Taxonomy levels: remembering, understanding, applying, analyzing, evaluating, and creating.
- Flashcards must include a question, answer, difficulty level, and topic tag.
- Reinforcement Learning from Human Feedback (RLHF) must be used to refine flashcard style and difficulty based on student interactions.

4. Voice Note Creation

- Convert lecture summaries into audio using Text-to-Speech (TTS).
- Provide two delivery modes: revision mode (short key points) and concept mode (detailed explanations).

- Voice notes must be clear, accurately pronounced, and mapped to corresponding syllabus topics.

5. Continuous Assessment (CA) Guidance and References

- Implement an Agentic Retrieval-Augmented Generation (RAG) pipeline supported by Knowledge Graphs to provide step-by-step solutions for CA problems.
- Include verified references such as textbooks, academic papers, or official documentation for each CA solution.
- Ensure guidance is syllabus-specific and adapts to institutional question formats.

6. Syllabus Mapping and Output Management

- Use a Knowledge Graph to ensure all outputs (summaries, flashcards, voice notes, CA guidance) are mapped to syllabus concepts and dependencies.
- Allow students to export outputs in multiple formats including PDF, CSV, audio files, and interactive flashcards.

7. User Interaction

- Users (students) must be able to query specific topics or CA questions.
- Provide an interactive web interface for navigating summaries, flashcards, and voice notes.
- Enable feedback collection on outputs (helpfulness, clarity, difficulty), which feeds into RLHF for continuous improvement.

3.7.2 Non-Functional Requirements

1. Performance

- Summarization and flashcard generation should process lecture slides efficiently, producing outputs within a reasonable time frame.

- Continuous Assessment (CA) guidance responses should be generated in near real-time to support active learning.

2. Scalability

- The system must handle multiple lecture files or large datasets simultaneously without performance degradation.
- The architecture should support deployment with containerization and orchestration tools such as Docker and Kubernetes for scalability.
- The system should be adaptable for use in other academic modules beyond DMS.

3. Reliability and Accuracy

- Summaries and CA guidance must maintain high fidelity with lecture content and syllabus, verified against a Knowledge Graph.
- Flashcards must accurately reflect syllabus concepts and factual correctness while adapting difficulty through RLHF.

4. Usability

- The system should provide a simple, intuitive interface for students to access summaries, voice notes, and flashcards.
- Search, filtering, and navigation options must be available for syllabus topics and CA questions.
- Multimodal access should ensure that students can learn via text, audio, and interactive flashcards.

5. Security and Privacy

- Institution-specific CA datasets and lecture materials must be securely protected against unauthorized access.

- User interactions and feedback logs should be stored securely in compliance with academic privacy policies.

6. Maintainability and Extensibility

- A modular architecture must be followed to enable upgrading of LLMs, RAG pipelines, and TTS engines.
- New features such as interactive simulations or video-based content should be easily integrated.
- Feedback mechanisms should ensure continuous improvement of summaries and flashcards through RLHF.

3.8 Testing

Testing plays an essential role in ensuring that the proposed system performs reliably and produces accurate, syllabus-aligned learning materials. The process is carried out in several stages, each focusing on verifying different aspects of the component.

- **Unit Testing**: Each module including data preprocessing, LLM-based summarization, flashcard generation, Text-to-Speech (TTS), Knowledge Graph integration, and CA guidance, is tested independently. For example, unit tests confirm whether lecture slides are parsed accurately, summaries align with syllabus concepts, and flashcards are generated according to Bloom's Taxonomy levels.
- **Integration Testing**: After individual modules are verified, they are combined to ensure smooth interaction. This stage confirms whether outputs from the summarization pipeline flow correctly into the flashcard generator, whether RLHF-based feedback is captured for adaptation, and whether the TTS module reliably converts the same summaries into audio.
- **System Testing**: The entire system is tested end-to-end to validate both functional and non-functional requirements. This includes verifying the accuracy and syllabus alignment of

summaries, the adaptivity of Bloom's-based flashcards, the clarity of voice notes, and the responsiveness of Agentic RAG–driven CA guidance. Reliability, scalability, and performance across devices and browsers are also evaluated.

- **User Acceptance Testing (UAT)**: The system is tested with a sample group of students and lecturers to ensure that it meets academic needs and syllabus standards. Special attention is given to evaluating syllabus coverage using the Knowledge Graph and adaptivity through RLHF. Feedback from this phase is used to refine and optimize the solution before deployment.

Through these testing stages, the project ensures that the system is dependable, user-friendly, and effective in supporting student learning.

3.9 Timeline

Task	Start Date	End Date	Duration
Feasibility Study			
Identify the Opportunity	May 2025	May 2025	1 week
Evaluate Feasibility & Background Study	May 2025	May 2025	1 week
Identify the Market Details	May 2025	May 2025	1 week
Requirement Gathering & Prioritization			
Requirement Gathering	May 2025	Jun 2025	2 weeks
Background Survey	May 2025	Jun 2025	2 weeks
Literature Review	May 2025	Jun 2025	2 weeks
Requirement Analysis	Jun 2025	Jul 2025	3 weeks
Software Requirement Specification	Jul 2025	Aug 2025	3 weeks
Identifying Functional Requirements	Jul 2025	Aug 2025	3 weeks
Identifying Non-Functional Requirements	Jul 2025	Aug 2025	3 weeks
Proposal			
Project Proposal Document	Aug 2025	Aug 2025	1 week
Project Proposal Presentation	Aug 2025	Aug 2025	1 week
Software Design			
Architectural Design	Sep 2025	Sep 2025	3 weeks
Database Design	Sep 2025	Sep 2025	3 weeks
Designing Wireframes	Sep 2025	Oct 2025	4 weeks
Implementation			
Environment Setup	Oct 2025	Nov 2025	3 weeks
ML Component Development	Nov 2025	Dec 2025	4 weeks
Frontend Development	Dec 2025	Jan 2026	4 weeks
Software Integration	Jan 2026	Jan 2026	1 week
Software Testing			
Unit Testing	Jan 2026	Feb 2026	3 weeks
Integration Testing	Feb 2026	Feb 2026	1 week
System Testing	Feb 2026	Feb 2026	2 weeks
Acceptance Testing	Feb 2026	Feb 2026	2 weeks
Deployment & Maintenance			
Software Hosting	Feb 2026	Feb 2026	1 week
Software Maintenance	Feb 2026	Ongoing	Ongoing

Progress Presentations			
Progress Presentation I	Jun 2025	Jun 2025	1 week
Progress Presentation II	Sep 2025	Sep 2025	1 week
Final Documentation & Presentation			
Final Documentation	Jan 2026	Jan 2026	2 weeks
Final Presentation & Viva	May 2026	May 2026	1 week

Table 3- Project Timeline

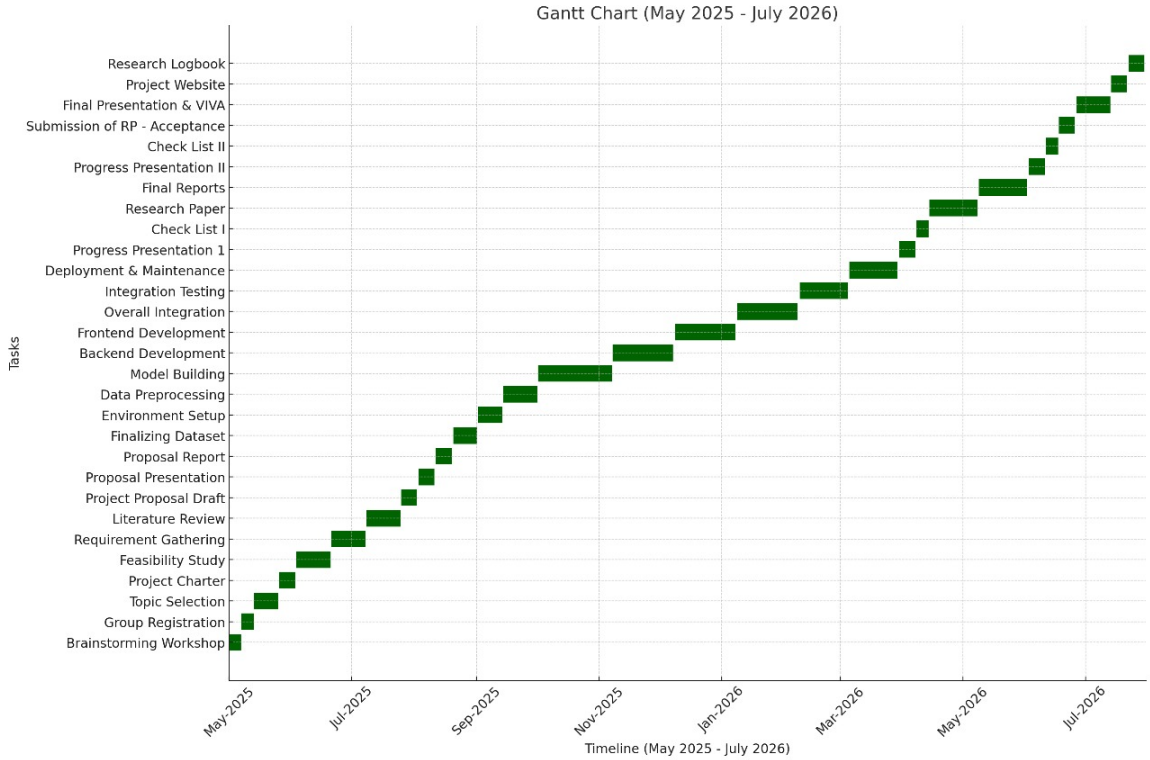

Figure 5- Project Grantt Chart

3.10 Risk Management Plan

3.10.1 Introduction

Risk management is essential to ensure the success of the academic support system. The main focus is to identify risks such as data quality issues in lecture slides and CA datasets, reliability challenges with Large Language Models (LLMs) and Retrieval-Augmented Fine-Tuning (RAFT), bias in Reinforcement Learning from Human Feedback (RLHF), and integration difficulties with Knowledge Graphs and Text-to-Speech (TTS). By applying early monitoring and mitigation strategies, the project can minimize these risks and maintain steady progress toward its objectives.

3.10.2 Risk Identification

Risk Category	Risk Description	Mitigation Strategies
Technical Risks	Inaccurate summaries or CA guidance due to challenges in fine-tuning LLMs, RAFT pipelines, or Agentic RAG models.	Conduct iterative testing, use syllabus-specific data, and consult domain experts.
	Limited computing resources affecting LLM, RLHF, or TTS processing.	Use optimized lightweight models, apply code efficiency practices, and scale with cloud resources if required.
Data-Related Risks	Inconsistent or misaligned lecture slides and CA datasets reducing output accuracy.	Source data from institution-approved materials, apply preprocessing and cleaning methods.
	Risk of outputs not mapping correctly to syllabus requirements.	Cross-check with subject lecturers and syllabus alignment checks.
Team Risks	Team member unavailability due to exams, illness, or commitments.	Apply cross-training, create contingency plans, and reallocate tasks when necessary.
	Miscommunication or coordination issues during module integration.	Use Jira for tracking, hold weekly meetings, and maintain documentation.
Supervisor Risks	Delayed feedback or decisions from supervisor/panel.	Set clear timelines, send reminders, and follow up proactively.

	Changes in scope or requirements during development.	Apply formal change control process, assess impacts, and negotiate adjustments.

Table 4- Risk Identification

3.10.3 Risk Monitoring and Control

Risks will be continuously monitored throughout the lifecycle of the academic support system. The project lead will oversee the tracking of identified risks and ensure that mitigation strategies are implemented effectively. Any new risks that arise, such as reliability challenges with LLMs, RAFT pipelines, RLHF feedback loops, or syllabus misalignment within the Knowledge Graph, will be documented and addressed promptly. Regular review meetings will be conducted to evaluate the status of each risk, assess the effectiveness of mitigation measures, and adjust the project plan where necessary to maintain steady progress and ensure alignment with academic objectives.

3.10.4 Conclusion

The success of this component depends heavily on proactive risk management. By identifying potential risks early, whether technical, data-related, or supervisory and establishing clear strategies to address them, the team can minimize disruptions and delays. Effective monitoring and timely mitigation will ensure that the system is delivered within scope, on schedule, and aligned with the academic expectations of both students and supervisors.

3.11 Communication Management Plan

The Communication Management Plan ensures that all stakeholders, including the supervisor, co-supervisor, and project team members, receive timely and accurate information regarding the progress of the system. Effective communication is vital for coordinating activities such as summarization, flashcard generation, TTS integration, and CA guidance development. This plan outlines the communication objectives, mediums, and practices followed throughout the project.

3.11.1 Communication Objectives

Communication within the project will aim to:

Inform: Share regular updates on progress, milestones, and challenges.

Persua-de: Gain approval or support for technical decisions, tool usage, or workflow changes.

Engage: Encourage collaboration among developers, supervisors, and testers.

Request: Seek inputs, clarifications, or resources from supervisors and team members.

Clarify: Resolve issues related to syllabus alignment, technical challenges, or testing outcomes.

3.11.2 Communication Mediums

The following communication mediums will be used to ensure effective coordination:

Email – Formal communication, progress reports, documentation, and record-keeping.

Phone Calls – Quick clarifications and immediate feedback on urgent issues.

WhatsApp Messaging – Informal, real-time updates and quick exchanges within the team.

Meetings – Group discussions for decision-making, reviewing progress, and resolving issues. These will be conducted weekly with the team and monthly with the supervisor.

Project Management Tools (Jira) – Tracking tasks, reporting issues, and monitoring sprint progress.

4. Commercialization

The proposed system serves as a valuable learning tool for students in the DMS module, offering automated lecture summarization, flashcards, voice notes, and continuous assessment guidance. To ensure accessibility, the platform can be provided as a free version with limited features, enabling students to experience the core functionalities without cost. Advertisements or institutional sponsorships may be used to support the free version, ensuring sustainability.

In addition, a premium subscription model can be introduced to provide advanced capabilities, such as access to a larger repository of CA guidance, enhanced flashcard sets, downloadable voice notes, and personalized study recommendations. This tiered approach allows students to choose between free basic tools and more advanced, premium features depending on their needs.

By focusing on the requirements of students preparing for demanding modules like DMS, the platform can significantly improve study efficiency, exam readiness, and overall learning outcomes. Through this dual model of free and premium access, the system remains inclusive while also ensuring a viable path for long-term growth and development.

5. Budget

A concise budget has been prepared to support the development and deployment of the lecture summarization and study support system. It covers essential expenses such as internet and hosting services, training, publication, and basic stationery. The estimated costs may vary depending on the final implementation requirements.

Types	Cost
Internet use and web hosting	LKR. 10,000.00
Training Cost	LKR.10,000.00
Publication Cost	LKR. 50,000.00
Stationery	LKR. 1,000.00
Additional	LKR. 10,000.00
Total	LKR. 81,000.00

Table 5- Budget for Component

6. Conclusion

In conclusion, the proposed system demonstrates its potential as a comprehensive academic support tool for students in the Database Management Systems (DMS) module by unifying multiple learning aids into a single, syllabus-aligned platform. Through the use of Large Language Models (LLMs) enhanced with Retrieval-Augmented Fine-Tuning (RAFT), the system generates concise and contextually accurate lecture summaries that simplify complex content. The inclusion of Bloom's Taxonomy–based flashcards, reinforced with Reinforcement Learning from Human Feedback (RLHF), strengthens active recall and adapts dynamically to student feedback. The Text-to-Speech (TTS) module supports multimodal learning by providing accessible voice notes, catering to diverse learner preferences. Furthermore, the integration of an Agentic Retrieval-Augmented Generation (RAG) pipeline, supported by Knowledge Graphs (KGs), ensures that continuous assessment tasks are addressed with step-by-step, syllabus-specific guidance and verified references.

By delivering these outputs in a structured, adaptive, and multimodal format, the system improves study efficiency, promotes conceptual clarity, and enhances knowledge retention. Its design reflects a balance between academic rigor and technological innovation, ensuring reliability, personalization, and inclusivity. Overall, this component contributes significantly toward strengthening exam preparation and creating an intelligent, feedback-driven learning environment tailored to the needs of DMS students.

7. References

[1] Bommasani, R., Hudson, D. A., Adeli, E., Altman, R., Arora, S., von Arx, S., … & Liang, P. (2021). On the opportunities and risks of foundation models. *arXiv preprint* arXiv:2108.07258.

[2] Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., … & Kiela, D. (2021). Retrieval-augmented generation for knowledge-intensive NLP tasks. *Advances in Neural Information Processing Systems, 34*, 9459–9474.

[3] Ouyang, L., Wu, J., Jiang, X., Almeida, D., Wainwright, C., Mishkin, P., … & Lowe, R. (2022). Training language models to follow instructions with human feedback. *arXiv preprint* arXiv:2203.02155. https://arxiv.org/abs/2203.02155

[4] Christiano, P. F., Leike, J., Brown, T., Martic, M., Legg, S., & Amodei, D. (2017). Deep reinforcement learning from human preferences. In *Advances in Neural Information Processing Systems* (pp. 4299–4307).

[5] Izacard, G., Lewis, P., Riedel, S., Karpukhin, V., Stoyanov, V., & Yih, W. T. (2022). Few-shot learning with retrieval-augmented language models. *arXiv preprint* arXiv:2208.03299. https://arxiv.org/abs/2208.03299

[6] Chen, J., Zhang, W., Liu, Y., Sun, J., & Zhao, X. (2023). KG-RAG Tutor: Knowledge graph enhanced retrieval-augmented generation for education. *arXiv preprint* arXiv:2311.17696. https://arxiv.org/abs/2311.17696

[7] Mayer, R. E. (2009/2021). *Multimedia Learning* (2nd/3rd ed.). Cambridge University Press.

[8] Anderson, L. W., & Krathwohl, D. R. (2001). *A taxonomy for learning, teaching, and assessing: A revision of Bloom's taxonomy of educational objectives*. Longman.

[9] Banerjee, A., Singh, R., Kumar, M., & Dutta, A. (2023). AI-enabled intelligent assistant for higher education: Generating quizzes, flashcards, and personalized guidance. *arXiv preprint* arXiv:2309.10892. https://arxiv.org/abs/2309.10892

[10] Nkambou, R., Bourdeau, J., & Mizoguchi, R. (Eds.). (2018). *Advances in Intelligent Tutoring Systems*. Springer.